angle between the neutron and the incident proton beam. A necessary requirement, however, is that the threshold energy ($1.64 \times (8/7) = 1.88$ MeV) must be exceeded, the Q-value for reaction (c) being -1.64 MeV.

11.4. NEUTRON CAPTURE AND SCATTERING

Unlike charged particles, no Coulomb barrier hinders neutrons from reaching the target nucleus. This leads to generally higher reaction cross-sections for neutrons, particularly at very low energies. Moreover, since neutrons can be produced in very high fluxes in nuclear reactors, neutron-induced processes are among the more important and well studied nuclear reactions.

We have seen that the geometric cross-section of a target nucleus is in the order of 1 b, or 10^{-28} m^2. Experimentally, the cross-sections for capture of energetic ("fast") neutrons ($E_n \geq 1$ MeV) are often close to 1 b. However, for neutrons whose kinetic energy is in the $1-100$ eV region, some nuclei show very large cross-sections – as high as 10^5 b. Such values can be explained as being due to neutron capture where the compound nucleus is excited exactly to one of its discrete energy levels (resonance capture). This does not mean that the nucleus is larger than its calculated geometric cross-section but that the interaction probability is very large in such cases – greater than the calculation of πr^2 would indicate. This can be understood as a case where sometimes reaction occur between the tails of the overlapping wave functions even when the overlap is small.

For low energy ("slow") neutrons ($E_n \ll 1$ MeV) the cross-section is also larger than the πr^2 value, and decreases as the velocity increases; this relation, $\sigma \propto v^{-1}$, is shown for boron in Figure 11.4. The relationship between the cross-section and the neutron velocity can be understood in wave mechanical terms since the wavelength associated with the neutron increases with a decrease in velocity. According to the matter-wave hypothesis (§2.3) the wavelength associated with a moving particle is $\lambda = \hbar/(mv)$ which can be written:

$$\lambda = \hbar(2mE_{kin})^{-1/2} = 0.286 \times 10^{-8}(mE_{kin})^{-1/2} \tag{11.11}$$

While the wavelength of a slow neutron is about 0.1 nm, that of a fast neutron is less than 1/1000 of that. Since the reaction probability increases with increasing particle wavelength (lager wave function overlap), a slow-moving particle has a higher probability of reaction than a faster one of the same kind.

In effect, the wave properties make neutrons appear much larger than their geometric size and increases the probability of interaction with the nucleus. From (11.9) it follows that

$$\sigma_{capt} \propto \lambda_n \propto E_n^{-1/2} \propto v_n^{-1} \tag{11.12}$$

where λ_n, E_n, and v_n are the wavelength, kinetic energy, and velocity, respectively, of the neutron. This relation is known as the $1/v$ law. This law is valid only where no resonance

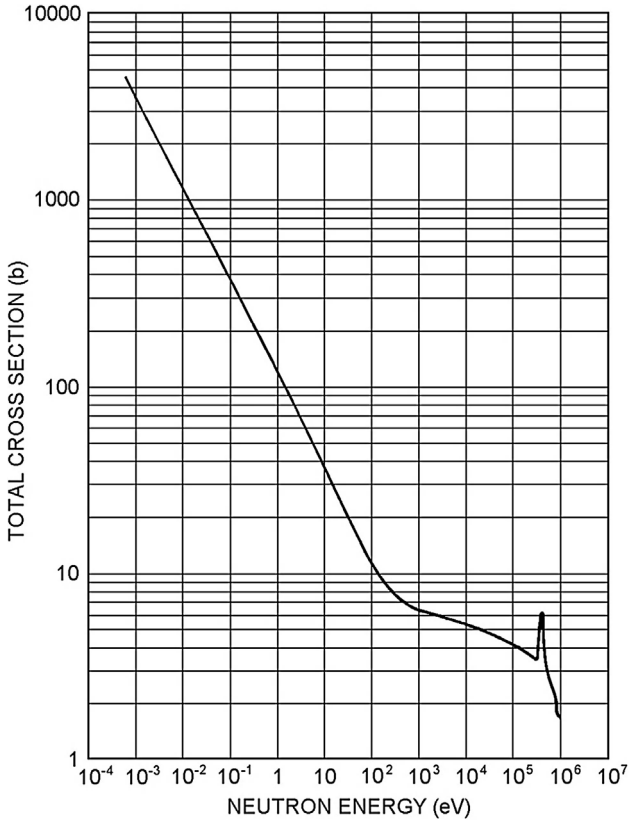


Figure 11.4 The total reaction cross-section of ^{10}B for neutrons of various kinetic energies.

absorption occurs. In Figure 11.4 $\sigma_{tot} = \sigma_{capt} + \sigma_{scat}$, but σ_{scat} is approximately constant. Since σ_{capt} decreases as E_n increases, at higher energies ($\geq$ 500 eV) σ_{scat} dominates over σ_{capt}, except for the resonance at 0.3 MeV. The capture or neutrons in ^{10}B leads to the formation of ^{7}Li and ^{4}He.

11.5. NEUTRON DIFFRACTION

From (11.12) we calculated that for neutrons the wavelength at thermal energies is on the order of 0.1 nm, i.e. of the same order of magnitude as the distance between atomic planes in a crystal. Thermal neutrons can, therefore, be scattered by crystals in the same manner as X-rays. For studies of crystal structures by neutron diffraction, a beam of mono-energetic neutrons can be obtained from the spectrum of neutron energies in a reactor by the use of monochromators.[1]

[1] Another method is mechanical time-of-flight based selectors.